Lecture 5: Distributions: Joint, Conditional and Classical

- This lecture first looks at situations with two or more random variables.
- Coverage includes joint probability distributions; joint mass and density functions; marginal distributions and conditional distributions.
- Make sure you know the definition of mean, variance of random variables and that of the covariance/correlation between two random variables. In discrete and continuous cases.

- Next, we cover some useful probability distributions that provide the building blocks for many statistical models.
- We start with some distributions for modelling discrete data, and then move on to some continuous distributions.
- We also look at evaluating probabilities for these distributions using R.

Joint Probability Distributions

- Often necessary to consider circumstances involving two or more random variables.
- If these are independent (an idea we define formally later) then sufficient to look at them separately.
- If they are associated, then need to work with probabilities involving all variables together.
- These probabilities are given by the **joint probability mass function** for discrete random variables, and **joint probability density function** for continuous random variables.

Joint Probability Mass Function

- The **joint probability function** f gives the probability of each pair of possible values for (X, Y).
  - Write these outcomes (x_i, y_j) for $i = 1, \ldots, k_x$ and $j = 1, \ldots, k_y$.
- Hence $\mathrm{P}(X = x, Y = y) = f(x, y)$.
- Can be useful to tabulate the probabilities.

	Value of Y			
Value of X	y_1	y_2	$\ldots$	y_{k_y}
x_1	$f(x_1, y_1)$	$f(x_1, y_2)$	$\ldots$	$f(x_1, y_{k_y})$
x_2	$f(x_2, y_1)$	$f(x_2, y_2)$	$\ldots$	$f(x_2, y_{k_y})$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
x_{k_x}	$f(x_{k_x}, y_1)$	$f(x_{k_x}, y_2)$	$\ldots$	$f(x_{k_x}, y_{k_y})$

- Note that $\sum_i \sum_j f(x_i, y_j) = 1$.

Marginal Probability Mass Functions

- The **marginal probability mass function** for X is obtained by summing along rows: i.e.

$$f(x_i) = \sum_j f(x_i, y_j)$$

- This describes the **marginal distribution** of X.
- This is simply the distribution we would have used for X had we never considered Y.
- Similarly, the marginal probability mass function for Y is given by summing down columns: i.e.

$$f(y_j) = \sum_i f(x_i, y_j)$$

Conditional Probability Mass Functions

- Suppose that we know that $X = x$.
- What is the probability mass function for Y conditional on this information?
- Defining events $A = \{Y = y\}$ and $B = \{X = x\}$, note that

$$\mathrm{P}(Y = y|X = x) = \mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{f(x, y)}{f(x)}$$

- Hence the **conditional probability mass function** of Y given that $X = x$ is defined by

$$f(y|x) = \frac{f(x, y)}{f(x)}.$$

- This defines the **conditional distribution** of Y given X.
- Similarly, the conditional probability mass function of X given $Y = y$ is defined

$$f(x|y) = \frac{f(x, y)}{f(y)}.$$

Independent Discrete Random Variables

- The random variables X and Y are **independent** if the events $A = \{X = x\}$ and $B = \{Y = y\}$ are independent for all possible pairs of outcomes (x, y).
- This occurs when $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$.
- Therefore in terms of mass functions, X and Y are independent if

$$f(x, y) = f(x)f(y)$$

  for all possible pairs of outcomes (x, y).
- Equivalently X and Y are independent if $f(x|y) = f(x)$ for all x, y.
- Very intuitive: X and Y are independent if knowledge of the value of Y tells us nothing about X.

Example: Illustrating joint distributions

Random variables X and Y have joint distribution defined by the following joint probability mass function:

	Y value		
X value	0	1	2
0	0.4	0.2	0.2
1	0.1	?	0.0

Questions:

(i) What is the value of $f(1,1)$?

(ii) What is the marginal distribution of X? and of Y?

(iii) Find the conditional distribution of Y given $X = 0$.

(iv) Are X and Y independent?

Example continued

Solutions:

(i) Since the probabilities must sum to 1, so

$$f(1,1) = 1 - (0.4 + 0.2 + 0.2 + 0.1 + 0.0) = 0.1$$

(ii) Summing along rows in the table, the marginal probability mass function for X is given by

X value	0	1
Probability	0.8	0.2

Similarly, summing down columns gives the marginal probability mass function for Y:

Y value	0	1	2
Probability	0.5	0.3	0.2

Example continued

Solutions continued

(iii) The conditional distribution of Y given $X = 0$ is given by the first row in the table of joint probabilities, all divided by $P(X = 0) = 0.8$. Hence the required conditional probability mass function is

Y value	0	1	2
Probability	0.5	0.25	0.25

(iv) We have

$$P(Y = 1|X = 0) = 0.25 \neq 0.3 = P(Y = 1)$$

Hence X and Y are not independent.

Situations with Three or More Random Variables

- Previous ideas extend naturally to joint distribution of three or more discrete random variables.
- For example, the probability mass function for the triplet X, Y, Z would be defined by $f(x, y, z) = \mathrm{P}(X = x, Y = y, Z = z)$.
- Can define a variety of marginal and conditional distributions.
  - E.g. $f(x, y|z) = \mathrm{P}(X = x, Y = y|Z = z)$ gives joint probabilities for X and Y conditional on Z.
- Often denote n random variables using subscripts: $X_1, X_2, \ldots, X_n$.
- Joint mass function is $f(x_1, x_2, x_3, \ldots, x_n)$.
- If independent, then

$$f(x_1, x_2, x_3, \ldots, x_n) = f(x_1)f(x_2)\cdots f(x_n) = \prod_{i=1}^{n} f(x_i)$$

Joint Continuous Distributions

- Previous ideas also transfer over to joint distributions of continuous random variables ('joint continuous distribution').
- Technical details are a little more complicated.

Joint probability density function

The distribution of a pair of continuous random variables X, Y, is described by their joint probability density function $f(x, y)$, for which

$$\mathrm{P}(a < X < b, c < Y < d) = \int_a^b \int_c^d f(x, y)\, dxdy$$

for any values a, b, c, d.

Marginal and Conditional Density Functions

- Can define the **marginal (probability) density function** of X by

$$f(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

- Often referred to as 'integrating out y'.
- Requires multi-variable calculus – or a good computing package!
- **Conditional (probability) density function** of X given Y given by

$$f(x|y) = \frac{f(x, y)}{f(y)}$$

- Marginal and conditional densities $f(y)$ and $f(y|x)$ defined analogously.
- Ideas extend to three of more variables in a similar way to the discrete case.

Independent Continuous Random Variables

- Continuous random variables X and Y are independent if and only if $f(x, y) = f(x)f(y)$.
- More generally, the n continuous random variables $X_1, X_2, \ldots, X_n$ are independent if and only if

$$f(x_1, x_2, \ldots, x_n) = f(x_1)f(x_2)\cdots f(x_n) = \prod_{i=1}^{n} f(x_i).$$

- We will revisit this material when we look at likelihood methods and Bayesian statistics in later parts of these notes.

Bernoulli Distribution

- Let random variable Y have a Bernoulli distribution with parameter p.
- For short, write $Y \sim \text{Bern}(p)$.
- Then Y can take values 0, 1 only.
- $\text{P}(Y = 1) = p$ and $\text{P}(Y = 0) = 1 - p$.
- The probability mass function of the Bernoulli distribution is

$$f(y) = f(y \mid p) = p^y (1 - p)^{1-y} \qquad y = 0, 1$$

- $\mathbb{E}[Y] = p$ and $\text{Var}(Y) = p(1 - p)$.
- Models 'biased coin toss' or any other binary response scenario (e.g. win/lose, rain/no rain).

Binomial Distribution

- Let $X_1, \ldots, X_n$ be independent $\mathrm{Bern}(p)$ random variables.
- Then $Y = \sum_{i=1}^{n} X_i$ is the total number of 'successes' (i.e. trials with a 1 outcome).
  - So Y can take values $0, 1, 2, \ldots, n$.
  - E.g. In coin tossing, $X_i = 1$ for head, $X_i = 0$ for tail, then Y total number of heads in n tosses.
- Random variable Y has a **binomial distribution** with parameters n and p.
- Typically denoted by $Y \sim \mathrm{Bin}(n, p)$.
- Probability mass function:

$$f(y \mid p) = \frac{n!}{y!(n-y)!} p^y (1-p)^{n-y} \quad y = 0, 1, \ldots, n$$

- $\mathbb{E}[Y] = np$ and $\mathrm{Var}(Y) = np(1-p)$.
- Note $\mathrm{Bin}(1, p)$ is same as $\mathrm{Bern}(p)$.

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Geometric distribution

- Let random variable Y have geometric distribution with parameter p.
- Write $Y \sim \mathrm{Geom}(p)$ for short.
- Y is number of failures before first success in a sequence of independent $\mathrm{Bern}(p)$ random variables.
- Hence Y can take values $0, 1, 2, \ldots$.
- Probability mass function:

$$f(y \mid p) = (1-p)^y p \qquad y = 0, 1, 2, \ldots.$$

- Mean and variance given by

$$\mathbb{E}[Y] = \frac{(1-p)}{p} \qquad \mathrm{Var}[Y] = \frac{(1-p)}{p^2}.$$

Poisson distribution

- Let random variable Y have Poisson distribution with parameter λ.
- Write $Y \sim \text{Pois}(\lambda)$ for short.
- Model counts of events occurring at random with no obvious maximum.
  - Y takes values $0, 1, 2, \ldots$.
  - E.g. number of earthquakes in some period.
- Probability mass function:

$$f(y \mid \lambda) = \frac{\lambda^y e^{-\lambda}}{y!} \qquad y = 0, 1, 2, \ldots.$$

- Parameter λ describes average rate at which invents occur.
- $\mathbb{E}[Y] = \lambda$ and $\text{Var}[Y] = \lambda$.
- $\text{Pois}(\lambda)$ approximates $\text{Bin}(n, p)$ with $\lambda = np$ when n large, p small.

Normal distribution

- Let random variable Y have normal distribution with parameters μ, σ^2.
  - Also called the **Gaussian distribution**.
- Write $Y \sim \mathrm{N}(\mu, \sigma^2)$ for short.
- Y can take any real number: $(-\infty, \infty)$.
- Good model for physical characteristics (like people's heights and weights), describing measurement error in experiments etc.
- Probability density function:

$$f(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} \qquad -\infty < y < \infty.$$

- $\mathbb{E}[Y] = \mu$ and $\mathrm{Var}(Y) = \sigma^2$.

Exponential distribution

- Let random variable Y have an exponential distribution with parameter λ.
- Write $Y \sim \text{Exp}(\lambda)$ for short.
- Y is a continuous random variable taking only non-negative values: $[0, \infty)$.
- Useful model for waiting times.
- Probability density function:

$$f(y \mid \lambda) = \lambda e^{-\lambda y} \qquad 0 \leq y < \infty.$$

- Mean and variance given by

$$\mathbb{E}[Y] = \frac{1}{\lambda} \qquad \text{Var}[Y] = \frac{1}{\lambda^2}$$

Gamma distribution

- The gamma distribution is a generalization of the exponential distribution.
- Let random variable Y follow a gamma distribution with **shape parameter** α and **rate parameter** λ.
- Write $Y \sim \mathrm{Gam}(\alpha, \lambda)$ for short.
- Y is a continuous random variable taking only non-negative values: $[0, \infty)$.
- Probability density function:

$$f(y \mid \alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\lambda y} \qquad 0 \leq y < \infty$$

  - here Γ denotes the mathematical **Gamma function**.
- Mean and variance given by

$$\mathbb{E}[Y] = \frac{\alpha}{\lambda} \qquad \mathrm{Var}[Y] = \frac{\alpha}{\lambda^2}$$

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Evaluating Probability Functions in R

- R has a variety of functions for evaluating probabilities and related quantities associated with common distributions.
- The syntax is helpfully uniform.
- To compute a probability mass or density function, R function name formed by prefix `d` added to R name for distribution.

Distribution	R name	Distribution	R name
Binomial	`binom`	Exponential	`exp`
Geometric	`geom`	Gamma	`gamma`
Poisson	`pois`	Uniform	`unif`
Normal	`norm`		

- For example:
  - `dpois` computes mass function for Poisson distribution.
  - `dexp` evaluates density function for exponential distribution.

Example: Poisson probability mass function

- Using Poisson distribution to illustrate syntax.
- `dpois(x, lambda)` evaluates Poisson mass function at value `x` when parameter value is `lambda`.
  - `x` and `lambda` can be vectors, in which case mass function at i-th element of `x` will use the i-th element of `lambda` (recycled if necessary).
- An optional argument is to set `log=T`, which returns (natural) logarithm of mass function.

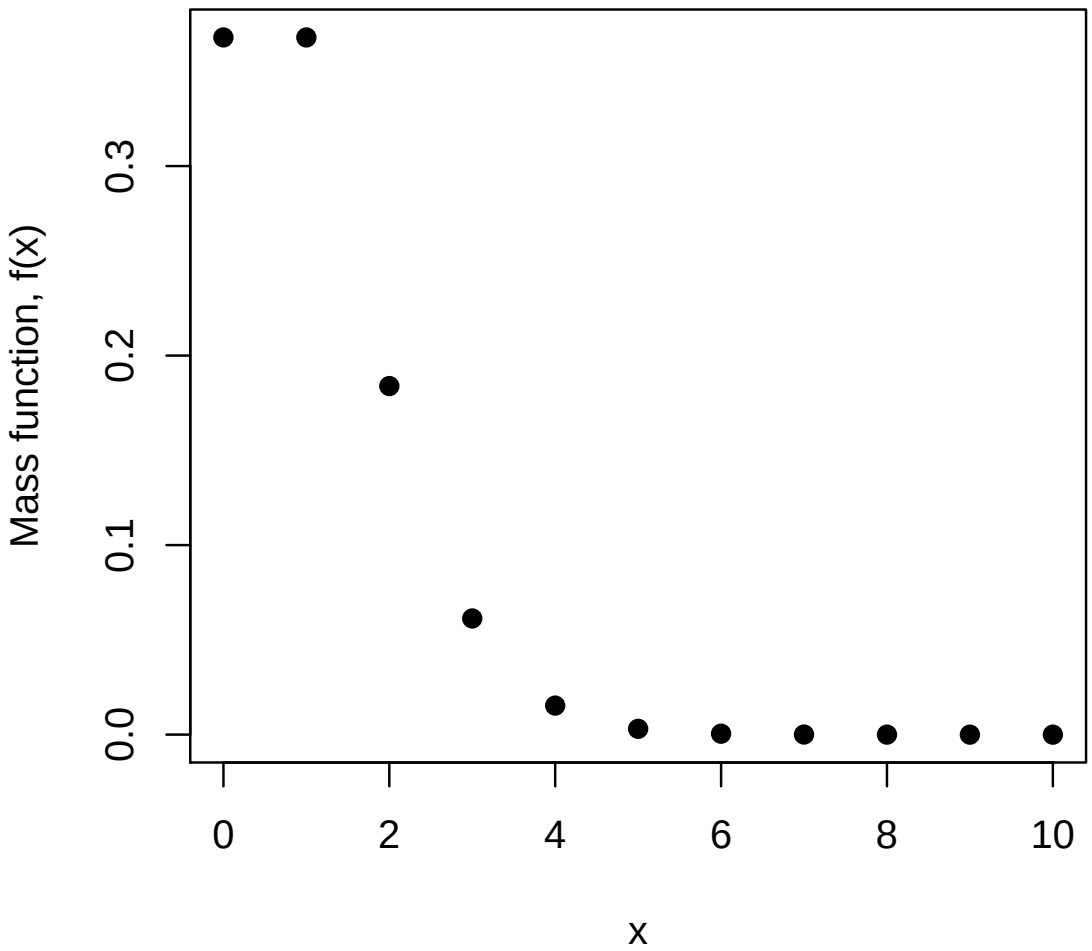
Example continued

R Snippet

```
> dpois(1,lambda=1)
[1] 0.3678794
> dpois(1,lambda=0.5)
[1] 0.3032653
> dpois(10,lambda=0.5)
[1] 1.632262e-10
> dpois(10,lambda=0.5,log=T)
[1] -22.53588
> dpois(c(1,3,5,7),lambda=c(1,2))
[1] 0.367879441 0.180447044 0.003065662 0.003437087
> x <- 0:10
> f <- dpois(x,lambda=1)
> plot(x, f, ylab="Mass function, f(x)", pch=19)
```

Example continued

Plotted figure



Example continued

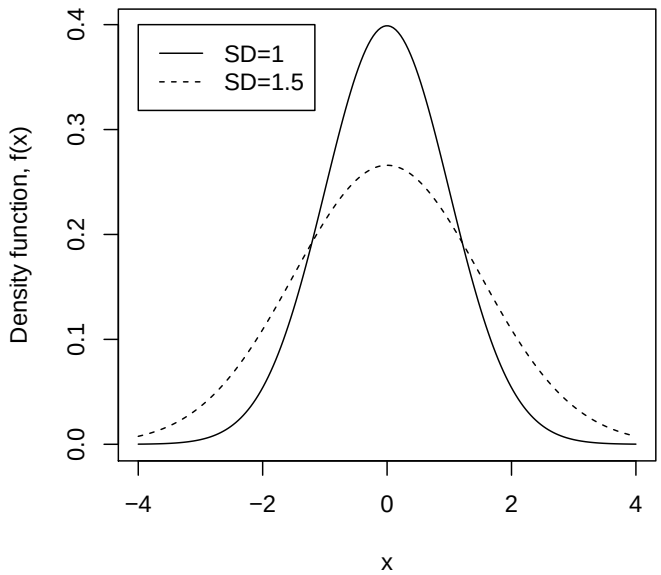
Some points to note

- In the command `dpois(c(1,3,5,7),lambda=c(1,2))` note that `lambda` is of length 2, and hence must be recycled to match length of the first argument.
- In the `plot` command:
  - `pch=19` sets the plotting symbol to a filled circle (rather than the default which is an unfilled circle).
  - `ylab` specifies the labelling for the y-axis.
  - x-axis labelling inherited from name of object, `x`.
- In plot, notice how quickly probabilities drop off.
  - With $\lambda = 1$, unlikely to see a Poisson random variable take values larger than 7.

Example: Plotting the Normal Density Function

R Snippet

```
> x <- seq(-4,4,length=200)
> f1 <- dnorm(x)
> f2 <- dnorm(x, sd=1.5)
> plot(x,f1,ylab="Density function, f(x)",type="l")
> lines(x,f2,lty=2)
> legend(-4,0.4,lty=c(1,2),legend=c("SD=1","SD=1.5"))
```



Example continued

Some points to note

- Density functions for $N(0, 1)$ and $N(0, 1.5^2)$ distributions plotted.
- Initial command sets up grid of x-values at which the densities will be evaluated.
- Objects `f1` and `f2` respectively contain density values for $N(0, 1)$ and $N(0, 1.5^2)$ distributions. `The plot`command takes the argument `type="l"` to indicate a line plot.
- Second density is added using the `lines` command, which adds lines to an existing plot.
- Argument `lty=2` sets line type, here to a dashed line (line type 2).
- `legend` command adds the legend to the plot.
  - Numbers specify position (x-y coordinates of top-left of legend box);
  - `lty` indicates that line segments of types specified be drawn,
  - `legend` argument supplies the text.

Calculating Cumulative Probabilities

- Probabilities for continuous distributions are given by areas under the density curves.
- For a random variable Y, the **cumulative distribution function** is defined by $F(y) = \mathrm{P}(Y \leq y)$.
- Probability for interval ranges for continuous distributions can be expressed using the cumulative distribution function.
  - E.g. $\mathrm{P}(a < Y \leq b) = F(b) - F(a)$.
- Implemented in R using functions `pdist`, where `dist` is replaced by appropriate distribution name.

R Snippet (Example of probability calculations)

```
> pnorm(0)
[1] 0.5
> pnorm(2,sd=1.5)
[1] 0.9087888
```